

DATA ENGINEERING COURSE

Introduction to Pandas

Assignment

Instructor: Tanishq Tyagi
Data Engineering Course

STUDENT INSTRUCTIONS

Instructions

- Read each question carefully before starting your solution.
- Write clean, well-commented Python code following PEP 8 guidelines.
- Use meaningful variable names throughout your code.
- Test your code with sample data before submitting.
- Each question must be answered in a separate code cell or clearly separated section.
- Ensure all import statements are included at the top of your solution file.
- Submit only your own original work.

ASSIGNMENT QUESTIONS

Assignment Questions

Question 1

Create a Series containing the names of five cities and assign a custom index to each entry.

Question 2

Read data from an Excel file that contains employee information, including at least the fields `employee_id` and `salary`, along with a few additional attributes of your choice.

Question 3

From the DataFrame loaded in Question 2, display only the first three rows using the appropriate pandas method.

Question 4

Use the `info()` method on the DataFrame to identify the data type of each column. Provide a brief written explanation of what the output tells you about the dataset.

Question 5

Use the `describe()` method to generate a statistical summary of the numeric columns in the DataFrame. For each statistic displayed in the output (count, mean, std, min, 25%, 50%, 75%, max), write a clear explanation of what it represents and why it is useful for understanding the dataset.